

DRY-RUN GUIDE | 01_ascending_order.cpp
Source: cpp\2nd_belt\01_ascending_order.cpp

CORE BEHAVIOR

Reads N values and checks each adjacent pair. Equal values are allowed; any decrease changes ascending to false and exits early.

REPRESENTATIVE INPUT / CALL

Input: N=4; values '1 2 2 5'.

STEP TRACE

Compare 2>=1, 2>=2, 5>=2; no decreasing pair occurs.

EXPECTED OUTPUT / RETURN

'Yes'

ACTUAL SOURCE EXCERPT

Verbatim lines from the source; long lines wrap. Source files were not modified.

```
001 | /*
002 | Question:
003 | You are given a list of integers. Your task is to check whether the list
004 | is in ascending order.
005 |
006 | A list is said to be in ascending order if every element is less than or
007 | equal to the next element.
008 |
009 | Print "Yes" if the list is in ascending order, otherwise print "No".
010 |
011 | Input:
012 | The first line contains an integer N - the number of elements in the list.
013 | The second line contains N space-separated integers.
014 | */
015 |
016 | #include <iostream>
017 | #include <vector>
018 | using namespace std;
019 |
020 | int main() {
021 |     int n;
022 |     cin >> n;
023 |
024 |     vector<int> arr(n);
025 |     for (int i = 0; i < n; i++) {
026 |         cin >> arr[i];
027 |     }
028 |
029 |     bool ascending = true;
030 |
031 |     for (int i = 1; i < n; i++) {
032 |         if (arr[i] < arr[i - 1]) {
033 |             ascending = false;
034 |             break;
035 |         }
036 |     }
037 | }
```

```
038 |     cout << (ascending ? "Yes" : "No");
039 |
040 |     return 0;
041 | }
042 |
```